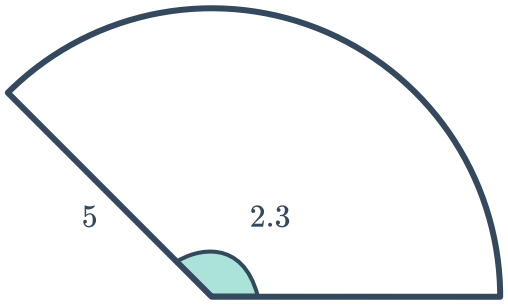
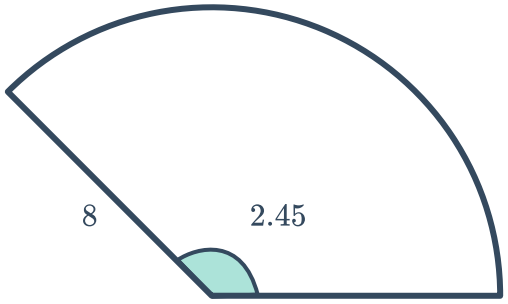
Arc length

1 Find the length of the arc of the following sectors, correct to one decimal place:

a



b



2 Find the length of the arc of the following sectors, correct to two decimal places:

a A sector of a circle of radius $4\ \text{cm}$, formed from an angle of size $\frac{5\pi}{6}$ radians.

b A sector of a circle of diameter 19 mm, formed from an angle of size 3.5 radians.

c A sector of a circle of radius $4\ \text{cm}$, formed from an angle of size $\frac{3\pi}{4}$ radians.

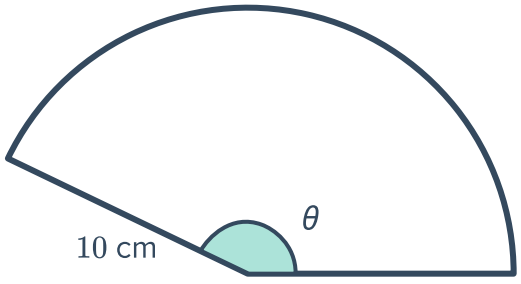
d A sector of a circle of radius $27\ \text{cm}$, formed from an angle of size $\frac{2}{9}$ radians.

3 The arc of a circle of radius $15\ \text{cm}$ subtends an angle of 150° at the center.

a Convert 150° to radians in exact form.

b Find the exact length of the arc, in terms of π.

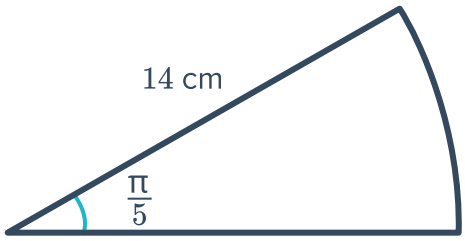
4 Consider the following sector of a circle, with radius $10\ \text{cm}$ and central angle θ equal to $\frac{5\pi}{6}$ radians:
Find the exact value of the perimeter of the sector.



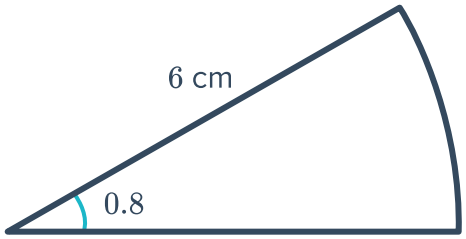
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5 Find the perimeter of the following sectors, correct to one decimal place:

a



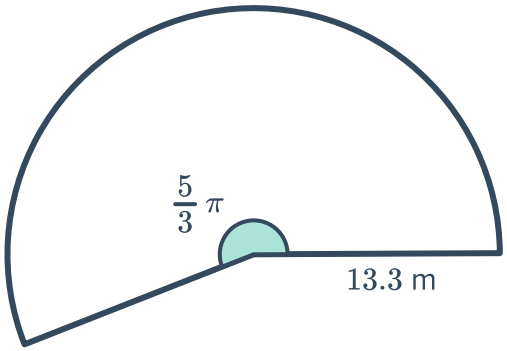
b



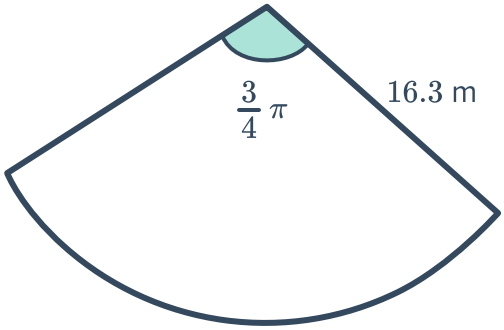
Sector area

6 Find the area of the following sectors, correct to two decimal places:

a



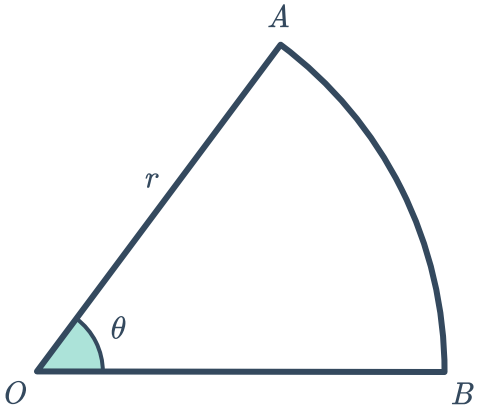
b



7 The diagram shows a sector with radius r and central angle θ:

The length of arc AB is $\frac{13\pi}{3}$ units.

Find the exact area of the sector when $r = 18$.



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8 Find the area of the following sectors. Round your answer to two decimal places where necessary:

a A sector of a circle with radius $5\ \text{cm}$, and subtended by an angle of 4 radians at the center.

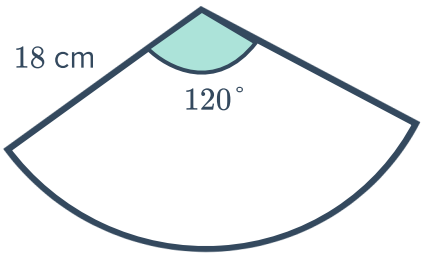
b A sector of a circle with a radius of $24.1\ \text{m}$ and a central angle of $\frac{4\pi}{5}$ radians.

c A sector of a circle with a radius of 22.8m and a central angle of $\frac{3\pi}{4}$ radians.

9 Consider the following sector:

a Convert the central angle of 120° to radians.

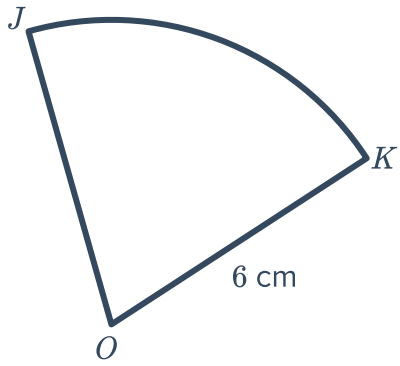
b Find the exact area of the sector in terms of π.



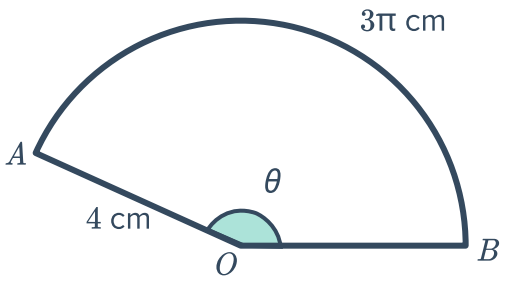
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10 Find the exact area of the following sectors in terms of π:

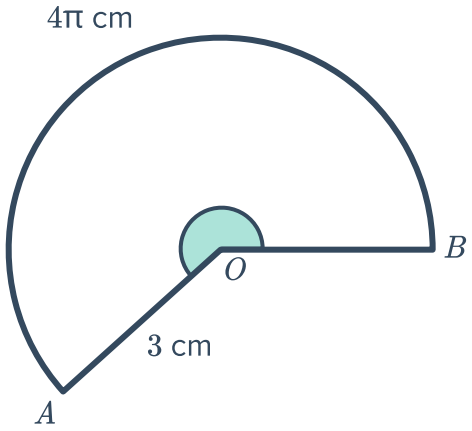
a The following sector has radius 6 cm and arc JK has length 2π cm:



b The following sector has radius 4 cm and arc AB has length 3π cm:



c The following sector has radius 3 cm and major arc AB has length 4π cm:

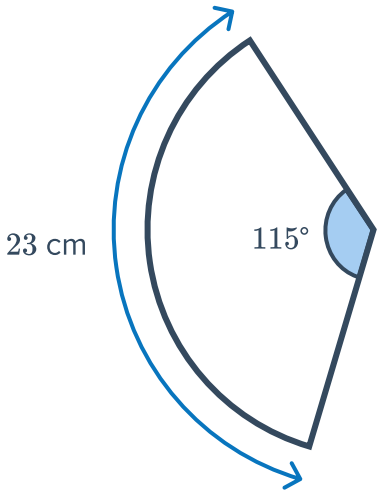


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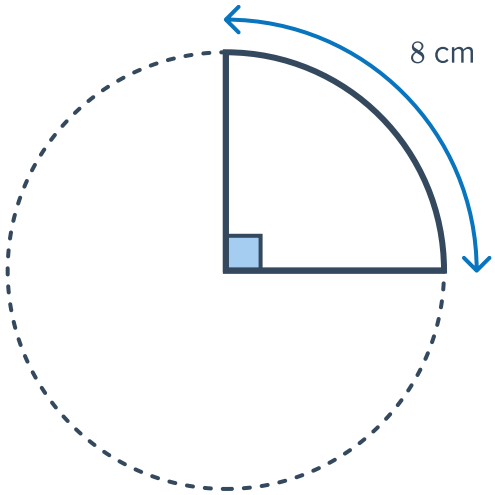
Dimensions from length and area

11 Find the radius of the following sectors to two decimal places:

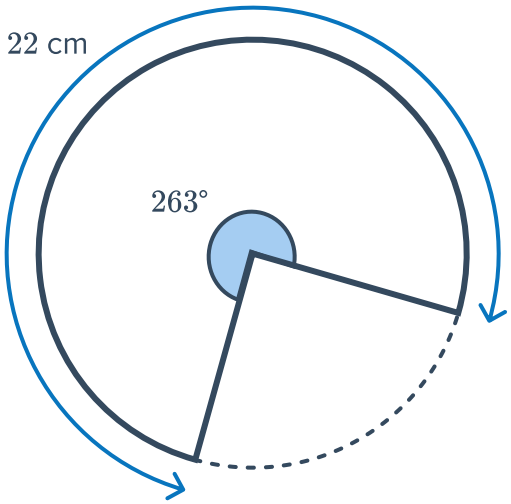
a



b



c

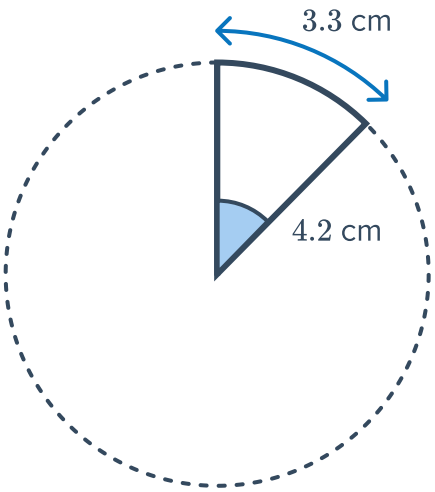


d A sector has a contained angle of 50° and an arc length of $13\ \text{cm}$.

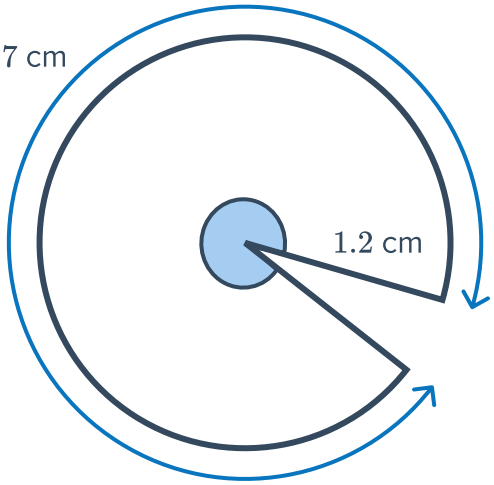
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12 Find the measure of the angle of the following sectors, in degrees. Round your answer to two decimal places.

a



b



c A sector with radius 4.2 cm and an arc length 14 cm.

13 A sector of a circle with radius 2 cm, has an arc length of 3.9 cm. Find the measure of the angle θ of the sector in radians, correct to two decimal places.

14 In each of the following, find the measure of θ, the angle subtended at the center of the circle, in radians:

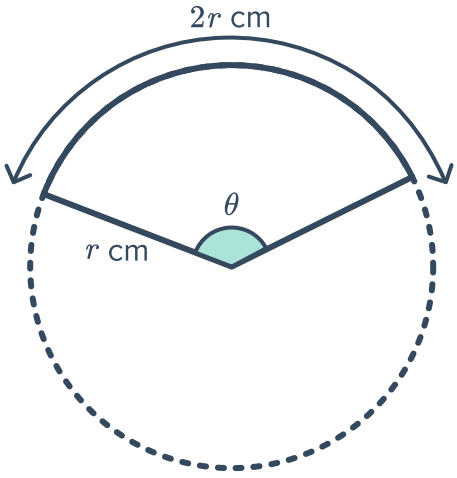
a The arc of a circle of radius 13 cm, measures 11.7 cm in length.

b The arc of a circle of radius 15 cm, measures 16.5 cm in length.

15 The diagram shows a sector of a circle with radius of r cm. The arc subtended by angle θ measures $2r$ cm in length.

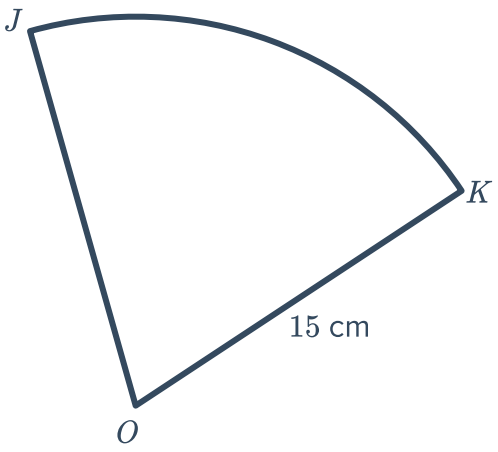
a Find the measure of the central angle θ, in radians.

b Find the exact area of the sector if $r = \frac{3}{2}\pi$ cm.



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16 The diagram shows an arc JK of a circle, with center O. The radius of the circle measures $15\ \text{cm}$ and the arc measures $12\ \text{cm}$ in length:



a Calculate the measure of $\angle JOK$ to the nearest degree.

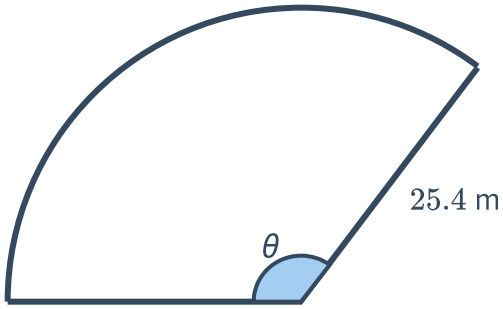
b Calculate the exact area of the sector.

17 A sector of a circle with radius $95\ \text{mm}$, has an arc length of $159\ \text{mm}$.

a Find the measure of the angle θ of the sector at the center of the circle in radians, correct to two decimal places.

b Convert the angle θ to degrees.

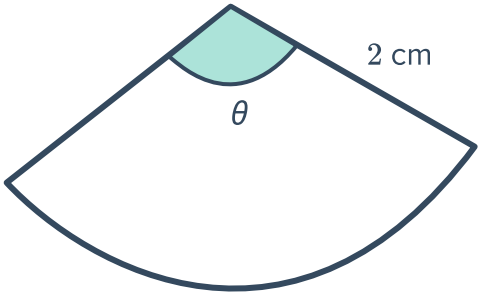
18 The perimeter of the following sector is $96.9\ \text{m}$:



a Use the formula $P = 2r + \frac{\theta}{180}\pi r$, to find the size of the angle θ, to the nearest degree.

b Find the area of the sector. Round your answer to the nearest integer.

19 The given sector has an area of $3.4\ \text{cm}^2$ and a radius of $2\ \text{cm}$:
Find the measure of θ in radians.



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20 A circle has a center at O and a radius r of $20\ \text{m}$. Sector OAC has an included angle of θ radians, and an arc length $l\ \text{m}$. The area of sector OAC is $160\ \text{m}^2$.

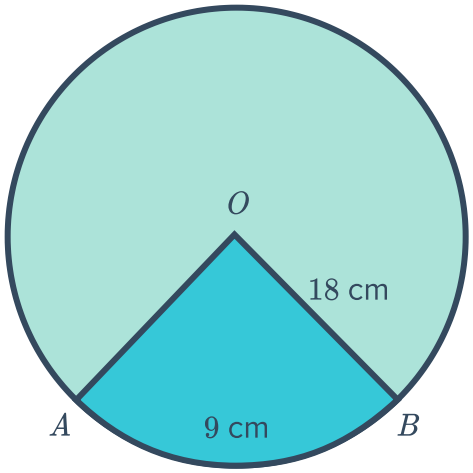
a Find θ, the measure of the angle $\angle AOC$ in radians.

b Find the value of l.

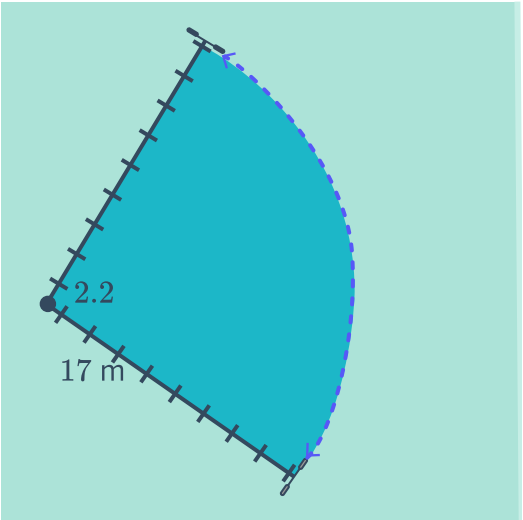
21 The area of a sector of a circle with a radius of $9\ \text{cm}$ is $49.5\ \text{cm}^2$. Find the length of the arc of the sector, l.

Applications

22 The given circle, centerd at O, has radius $18\ \text{cm}$ and arc AB has length $9\ \text{cm}$:
Find the area of the minor sector OAB.



23 A large $17\ \text{m}$ long sprinkler is placed in a crop field, with one end fixed and the other end free to move. As it turns, it waters everything underneath it.
If the sprinkler has turned an angle of 2.2 radians, find the area of the crop field it has watered.

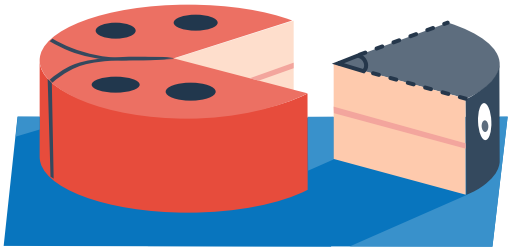


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24 Roald makes a lady beetle cake with a diameter of $24\ \text{cm}$ and a height of $10\ \text{cm}$. Roald removes a slice of the cake with an angle measure of $\frac{\pi}{4}$ radians.

a Find the exact area that the black icing covers on the removed slice (both the top and side).

b Find the exact area that the red icing covers on the remaining cake (both the top and side).

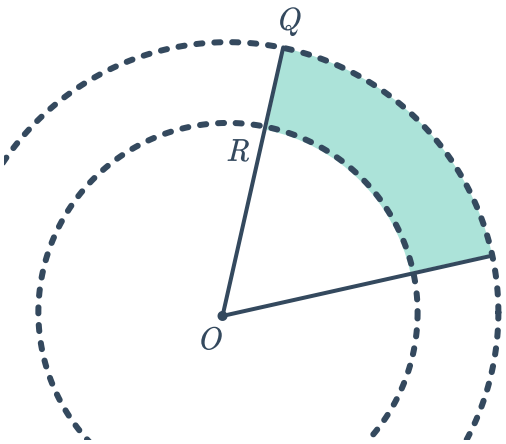


25 A sector formed from a central angle of $\frac{10}{9}\pi$ radians, is cut from a circular plastic sheet of radius $63\ \text{cm}$. The cut edges of the sector are brought together to form a plastic cone. Find the circumference of the circular base of the funnel, leaving your answer as an exact value in terms of π.

26 The diagram shows the sectors of two concentric circles with common center O where $\angle O = \frac{\pi}{4}$ radians, $OR = 8\ \text{cm}$, and $OQ = 12\ \text{cm}$.

a Calculate the exact area of the shaded region between the two circular arcs.

b Calculate the exact perimeter of the shaded region.



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